

Event Management System – Blood Donation Camp ER Model Documentation

Attributes

EVENT

Attribute	Type
Event_ID	Key Attribute (PK)
Event_Name	Simple
Event_Date	Stored
Venue	Composite (optional)
Organizer	Simple
Start_Time	Simple
End_Time	Simple
Status	Simple

DONOR

Attribute	Type
Donor_ID	Key Attribute (PK)
Name	Composite (First, Middle, Last)
DOB	Stored
Age	Derived from DOB
Gender	Simple
Blood_Group	Simple
Phone	Multivalued
Email	Simple
Address	Composite
Weight	Simple

REGISTRATION

Attribute	Type
Registration_ID	Key Attribute (PK)
Registration_Date	Stored
Registration_Status	Simple

APPOINTMENT

Attribute	Type
Appointment_ID	Key Attribute (PK)
Slot_Time	Simple
Queue_Number	Simple

MEDICAL_SCREENING

Attribute	Type
Screening_ID	Key Attribute (PK)
Blood_Pressure	Simple
Hemoglobin	Simple
Temperature	Simple
Eligibility	Derived/Simple
Remarks	Nullable

DONATION

Attribute	Type
Donation_ID	Key Attribute (PK)
Donation_Time	Stored
Quantity	Simple
Donation_Status	Simple

BLOOD_TEST

Attribute	Type
Test_ID	Key Attribute (PK)
HIV_Result	Simple
Hepatitis_Result	Simple
Malaria_Result	Simple
Overall_Result	Derived

STAFF

Attribute	Type
Staff_ID	Key Attribute (PK)
Name	Composite
Phone	Multivalued
Email	Simple
Role	Simple

EVENT_STAFF

Attribute	Type
Event_Staff_ID	Key Attribute (PK) or Composite
Duty	Simple

Attribute Classification

Key Attributes: Event_ID, Donor_ID, Registration_ID, Appointment_ID, Screening_ID, Donation_ID, Test_ID, Staff_ID, Event_Staff_ID.

Composite Attributes: Name (First, Middle, Last), Address (Street, City, State, PIN), Venue.

Multivalued Attributes: Phone.

Stored Attributes: DOB, Event_Date, Registration_Date, Donation_Time.

Derived Attributes: Age – DOB, Overall_Result – Test Results.

Null Attributes: Remarks, Middle_Name.

Structural Constraints

Relationship	Name	Cardinality	Participation
Donor-Registration	Registers	1:N	Partial
Event-Registration	Registers_For	1:N	Partial
Registration-Appointment	Assigned	1:1	Total
Donor-Medical_Screening	Undergoes	1:N	Partial
Event-Medical_Screening	Conducts	1:N	Total
Medical_Screening-Donation	Approves	1:1	Partial
Donor-Donation	Donates	1:N	Partial
Event-Donation	Occurs_At	1:N	Total
Donation-Blood_Test	Tested	1:1	Total

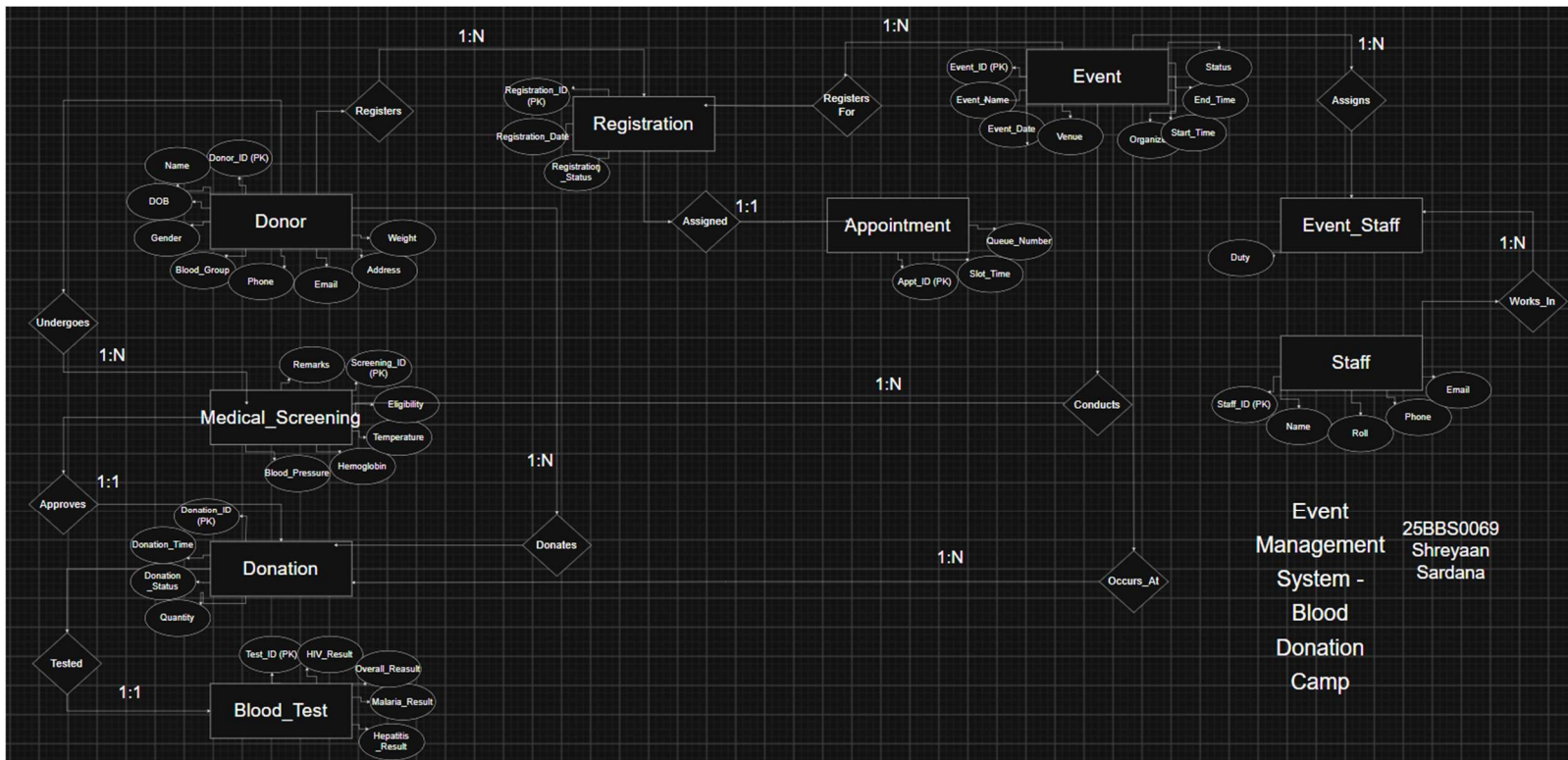
Event-Event_Staff	Assigns	1:N	Total
Staff-Event_Staff	Works_In	1:N	Partial

Degree of Relationships

Binary (Degree = 2): All relationships shown in the ER diagram.

Example Ternary (Degree = 3): DONOR – DONATES_AT – EVENT – supervised by STAFF.

ER Diagram



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